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Research Article

Design synthesis and crystallization of acetaminophen

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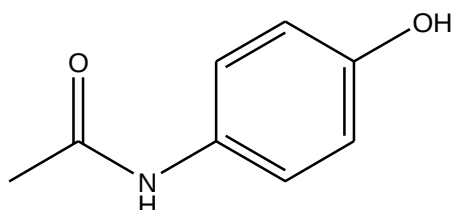
Abstract: Acetaminophen (paracetamol) is a synthetic non-opioid derivative of p-aminophenol and basic bioactive molecule in numerous pharmaceutical preparations for the treatment of colds and flu. In combination with opioid analgesics, acetaminophen can also be used in the management of more severe pain such as post-surgical pain and providing palliative care in advanced cancer patients. Design of synthesis of acetaminophen is based on a modern approach of choosing the right synthetic route and using methods necessary for the characterization of the resulting pharmaceutically active compound, thereby providing reliable parameters of the chemical quality of the synthesized molecules. Formation of the preferred form of acetaminophen is a process made up of two stages, where in the first stage an acid catalyzed reaction gives the crude acetaminophen. The reaction takes place according to the mechanism of nucleophilic addition in which the nitrogen atom of the p-aminophenol as nucleophile by attacking a carbonyl group of acetic anhydride forms an intermediate which undergoes the further elimination of the acetate anion. Recrystallization is one of the most common techniques used for purifying drug solids. One of the goals of this study was to achieve the results through the crystallization process in optimum conditions, in order to improve the yield and adequate purity of synthesized acetaminophen. In pharmaceuticals cyclodextrins have mainly been used as complexing agents to increase the aqueous solubility of poorly

water-soluble drugs through noncovalent inclusion complexation and to increase their bioavailability and stability. Complexation with beta cyclodextrin was attempted to improve solubility of acetaminophen.

Keywords: drugs, acetaminophen, synthesis, cyclodextrins.

INTRODUCTION

N-acylated aromatic amines (that have the acyl group $R-\overset{\text{O}}{\parallel}{C}-$ bonded to the nitrogen atom) such as phenacetin and acetaminophen are mild analgesics and antipyretics which can be obtained without a prescription. Acetaminophen is the basic bioactive molecule in a number of pharmaceutical preparations against colds and flu, such as Tylenol and Panadol (**Fig. 1**). Acetaminophen (paracetamol) or according to IUPAC N-(4-hydroxyphenyl) ethanamide belongs to the antipyretic non-opioid analgesic, molecular formula $C_8H_9NO_2$ and mass $151.18 \text{ g / mol}^{-1}$. The molecule consists of a benzene nucleus substituted by an hydroxyl group and the nitrogen atom of the amide group in the para (1,4) position ². The presence of two active groups also gives to the benzene nucleus high reactivity during electrophilic aromatic substitution.



N-(4-hydroxyphenyl) acetamide

Figure 1: Molecular structure of acetaminophen

pK of acetaminophen is 9.70 and is classified as a weak acid ³. Acetaminophen is present in the form of crystalline white powder, odorless, with a slightly bitter taste. The crystals formed from a saturated aqueous solution, have the form of monoclinic prisms. The concentrated aqueous solution of acetaminophen has a pH of 5.5 to 6.5. According to BCS classification system of drugs it belongs to the class III, which is characterized by high solubility and low permeability ⁴. The values of melting point of acetaminophen are in the range of $168-172^{\circ}\text{C}$. Acetaminophen is poorly soluble in cold water (1.4 g / 100 ml), in hot water solubility is higher and amounts to about 5 g/100 ml . It has a low solubility in non-polar (toluene, pentane, benzene, petroleum ether) and chlorinated hydrocarbons (CCl_4). It shows a significant solubility in solvents of moderate polarity N, N-dimethylformamide, alcohols and diethylamine⁵. It exists in five polymorphic forms (of which two can only be accessed at high pressure). Form I (**Fig 2.**) displays a herringbone arrangement of molecules within the crystal structure, whilst form II is layered ⁶.

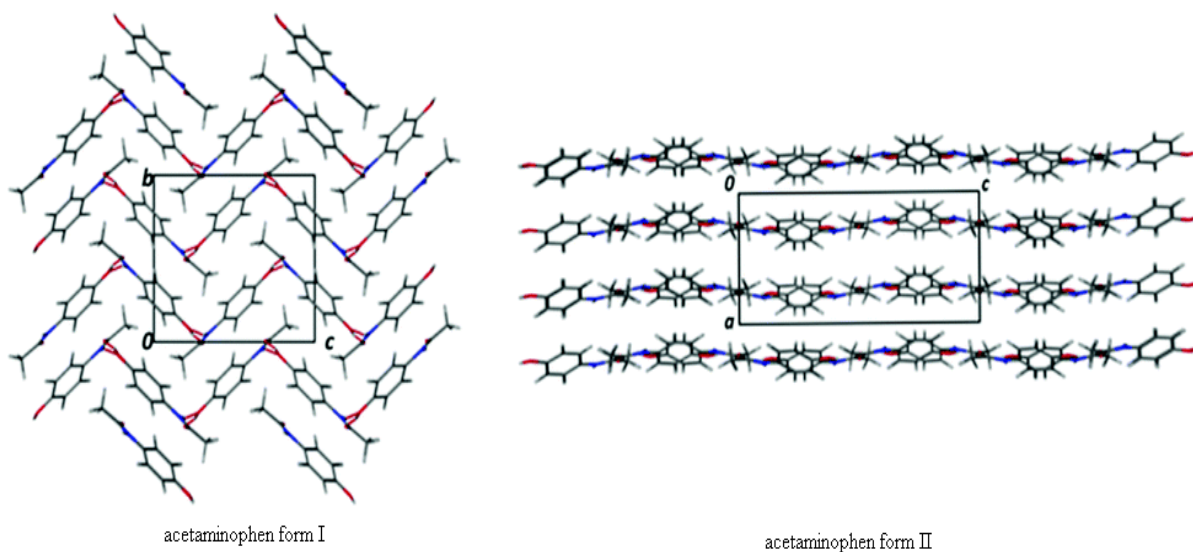


Figure 2: Polymorphic forms of acetaminophen.

Because of the commercial importance, we have paid the attention to the synthesis and purity of acetaminophen. The ideal synthesis of a compound would be the one that includes translating the most widespread and cheapest starting material in the desired molecule, including the smallest possible number of phases with the most accessible reagents and the highest possible total yield. The important stage in the manufacture of pharmaceutically active compounds is crystallization from a solution. Over 90% of all pharmaceutical products such as tablets, capsules, etc., include a pharmaceutically active substance in the solid state⁷. The overall success of the formulation process and the results of the mechanical properties of tablets, largely depend on the quality of crystal substances used during the tableting process.

The term crystallization usually involves the formation of the crystalline form of a particular solution⁸, whilst broader definition includes the precipitation process and the transformation of a solid state⁹. Methods of crystallization include the use of certain solvents, but different crystallisation conditions may lead to the occurrence of different crystal forms. We are constantly working on researches and finding new, innovative methods of manipulation of crystallization process¹⁰. Design of synthetic route of acetaminophen is based on four different laboratory methods of producing the crude acetaminophen by nucleophilic addition mechanism, followed by elimination.

Catalyzed by a mineral acid, acetic anhydride reacts with the amino group of p-aminophenol. In the first stage of the synthesis, the nucleophilic nitrogen atom of amino group of p-aminophenol attacks the carbonyl carbon atom of acetic anhydride towards nucleophilic addition mechanism; we get an intermediate compound which is subjected on further elimination of acetate anions (**Fig 3**). As a byproduct of the reaction acetic acid occurs from acetate anion and hydrogen proton which gained independence¹¹. The second phase is based on a process of recrystallization of the crude acetaminophen in order to get the product that meets the requirements of purity by using the appropriate solvent.

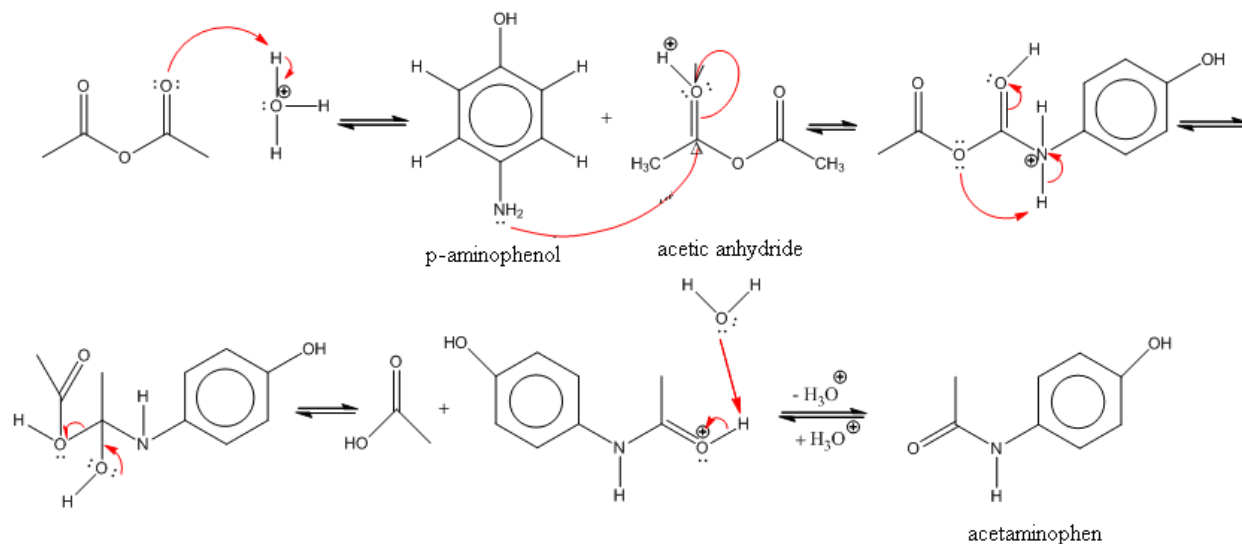


Figure 3: The reaction mechanism of acetaminophen synthesis

One of the main shortcomings of many drugs, which are usually very effective, is low solubility in water. Therefore, the therapeutic dose is higher than the effective dose which results in a reduced bioavailability of the drug in the body, by using oral therapy. The solution to this problem is the inclusion of the drug in the molecule with higher water solubility. Natural and hydrophilic derivatives of natural cyclodextrins are in the spotlight because of their role as complexing agents. Cyclodextrins as macrocyclic oligosaccharides usually consist of 6, 7 and 8 glucopyranose units which are characterized by the corresponding ring structure with appropriate ring diameter and the cavity within it ¹². Cyclodextrins are able to bind non-polar aliphatic and aromatic molecules of the same dimensions in a different ratio 1: 1, 2: 1 and 1: 2 (**Fig. 4.**), depending on the guest molecules ¹³.

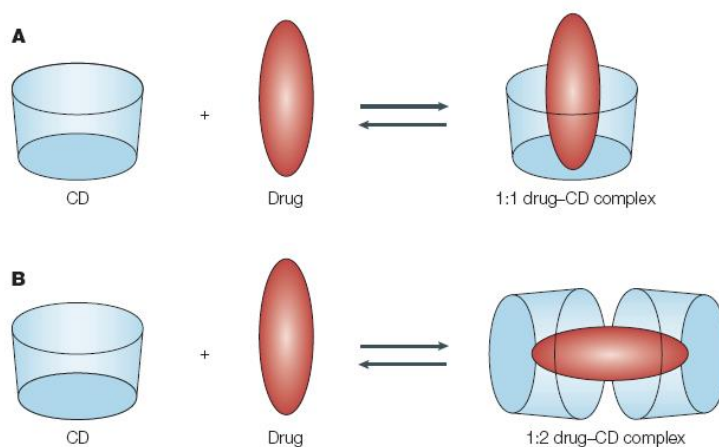


Figure 4: Drug complexation with cyclodextrin a) 1: 1 complex drug-CD, b) 1: 2 complex drug-CD.

MATERIAL AND METHODS

For a variety laboratory methods of acetaminophen synthesis were used the following substances:

- $\text{H}_2\text{NC}_6\text{H}_4\text{OH}$, 4-aminophenol (p-aminophenol), 97% (Acros)
- $\text{C}_4\text{H}_6\text{O}_3$, acetic anhydride, 99 % (Lach-Ner)
- H_3PO_4 , phosphorous acid, 85 % (Aldrich)
- Demineralized water, Distilled water, Ice.

To getting the crude acetaminophen, 4 (four) different synthetic routes were conducted for the purpose of comparison the same. The first method of synthesis is based on the following: weighed 1.5 g of p-aminophenol in the erlenmeyer flask and 25 ml of deionized water were added. Stirring is carried out with slow addition of concentrated phosphoric acid (H_3PO_4). The mixture was warmed over boiling water bath and 2 ml of acetic anhydride was added. The reaction mixture was warmed on a water bath for 20 minutes. After that, the mixture was cooled to room temperature and crystallization is carried out by additional cooling on ice bath for 20 minutes. The formed crystalline precipitate was filtered over a Büchner funnel, washed with a little cold distilled water and dried.

Second method of synthesis involved the same steps, only the time of mixture heating and cooling were different. The mixture was warmed on a water bath for 10 minutes and additional cooling and crystallization on the ice lasted 30 minutes. The third method of synthesis is different from the previous two because the round bottom flask for heating the mixture with a reflux condenser was used. Mixture warming time was 15 minutes and additional crystallization cooling lasted 20-30 minutes. Synthesis was performed without addition of concentrated H_3PO_4 . The fourth way is the most different from the previous three, because it didn't involve mixture heating and additional crystallization cooling. In glass weighed 1.5 g of p-aminophenol together with 25 ml of deionized water and acetic anhydride was added. The mixture was stand at room temperature for 30 minutes, and mixed periodically.

In the recrystallization of crude acetaminophen were used the following solvents:

- CH_3OH , methanol, 99,8% (Merck)
- $\text{C}_2\text{H}_5\text{OH}$, ethanol, 96% (Merck)
- $\text{C}_3\text{H}_7\text{OH}$, propanol, 99,8% (Merck)
- Aqueous solutions of methanol concentration 20%,40%,60%,80%.

Recrystallization of synthesized acetaminophen was performed in the manner that the crude product dissolved in appropriate amount of solvent to formed a supersaturated solution and then hot-filtered. Optimization of crystallization process where conducted at temperature of 6°C for 24 h and on room temperature. Deionised water was used in all the experiments. Complexation and characterization of synthesized samples were used by following chemicals: aqueous solutions of beta cyclodextrins different concentration (2-10 mM), potassium bromide, methanol, TLC eluent (55% ethyl acetate, 10% acetic acid and 35% petroleum ether).

For the purpose of analysis and characterization of synthesized acetaminophen, were applied following instrumental methods:

Fourier transform infrared spectroscopy (FTIR): FTIR spectra of acetaminophen were recorded on a Perkin-Elmer Spectrum 1000 system, equipped with a deuterium triglycine sulfate detector. The scan range was 350–4000 cm^{-1} , using eight scans per spectrum with a resolution of 4 cm^{-1} . Spectra were obtained in the transmission mode in KBr pellets.

Determination of melting point: Melting temperature of samples was determined by A.KRUSS Automatic Melting Point Meter - Semiauto Version, gradual warming, which includes a microscope for visual monitoring.

Ultraviolet Assay: An ultraviolet–visible (UV–Vis) spectrophotometer (Perkin-Elmer, Model Lambda 25), controlled by Perkin Elmer UV WinLab Software at the interface, was used to analyze complexation with beta cyclodextrins and acetaminophen solubility. Preliminary UV scanning of pure aqueous acetaminophen yielded a stable peak at 232 nm. A series of drug – cyclodextrin solutions was diluted as needed with respective solvents and analyzed by UV-Vis spectrophotometer. Fig. 5 show an excellent correlation ($r^2 > 0.9979$) indicated that the Beer–Lambert law was obeyed in the acetaminophen concentration ranges of interest. The solubility of acetaminophen in beta cyclodextrins solutions was measured at cyclodextrin concentrations between 0 and 10 mM. Excess amounts of acetaminophen were added to vials containing 10.0 mL of beta cyclodextrins solutions. The resulting suspension was treated at room temperature (25°C) with 150 rpm in an incubator orbital shaker for 72 hours. The solutions were allowed to equilibrate and analyzed on the basis of a pseudo-phase model ¹⁴. The samples were filtrated and the concentration of acetaminophen determined spectrophotometrically as described above. All the solubility experiments were carried out in triplicate.

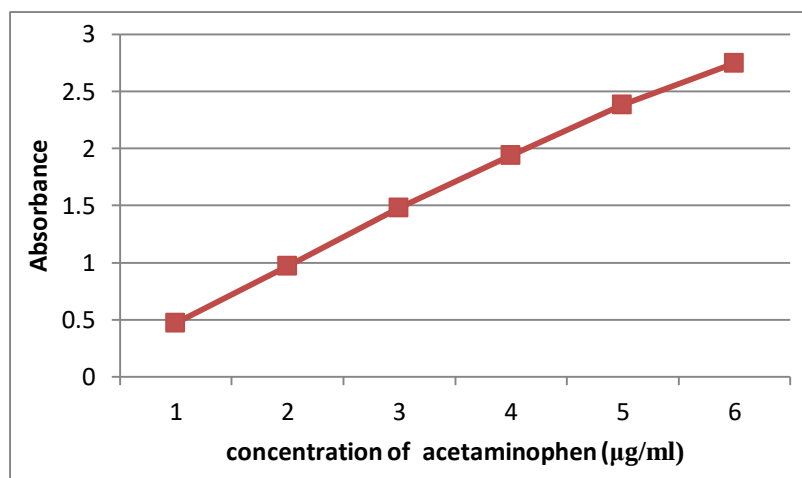


Figure 5: Calibration curve of acetaminophen in methanol

Thin layer chromatography (TLC): Thin layer chromatography is a method for preliminary synthetic testing, separation, identification and purity determination of various compounds ¹⁵. As stationary phase TLC analysis of acetaminophen samples were used silica gel plates (20x20). The mobile phase (eluent) containing 55% ethyl acetate, 10% acetic acid and 35% petroleum ether. The preparation is based on the dissolution of a small amount (1 mg) sample of p-aminophenol, crude (synthesized) and acetaminophen

samples paracetamol Ph.Eur. in methanol. Visualization of the sample was carrying out in Camag UV cabinet with a lamp at 254 nm. The obtained data were processed and ANOVA statistical data processing

RESULTS AND DISCUSSION

During the design of acetaminophen synthesis we started from four different methods that are based on the formation reaction of acetaminophen molecule from the starting molecules of 4-aminophenol and acetic acid (**Fig. 6**).

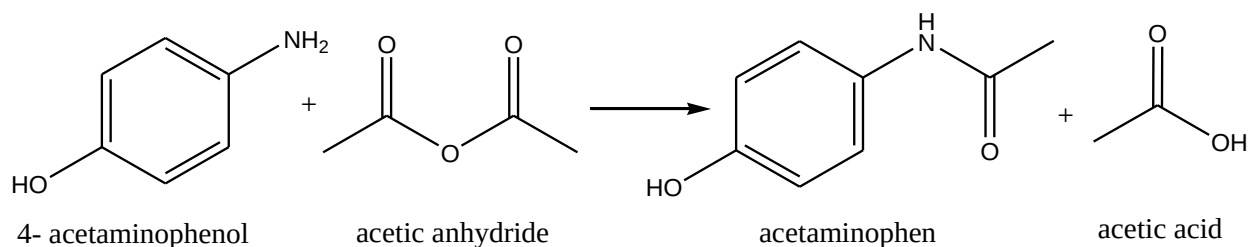


Figure 6: Synthetic pathway of acetaminophen

Figure 7 graphically shows the yield value of four different laboratory synthesis methods of acetaminophen. Actually, during the third method of synthesis the heating time of the reaction mixture using reflux was 15 minutes, in contrast to the first (I) method (20 minutes) and second (II) method (10 minutes). However, the key difference between the first and second method of synthesis is that in the design process of the third method we did not use phosphoric acid (H_3PO_4) as the initiator. A slightly lower yield of the fourth synthetic reaction method in relation to the third method of synthesis is based on the fact that the reaction is carried out at room temperature without additional cooling during the process of crystallization.

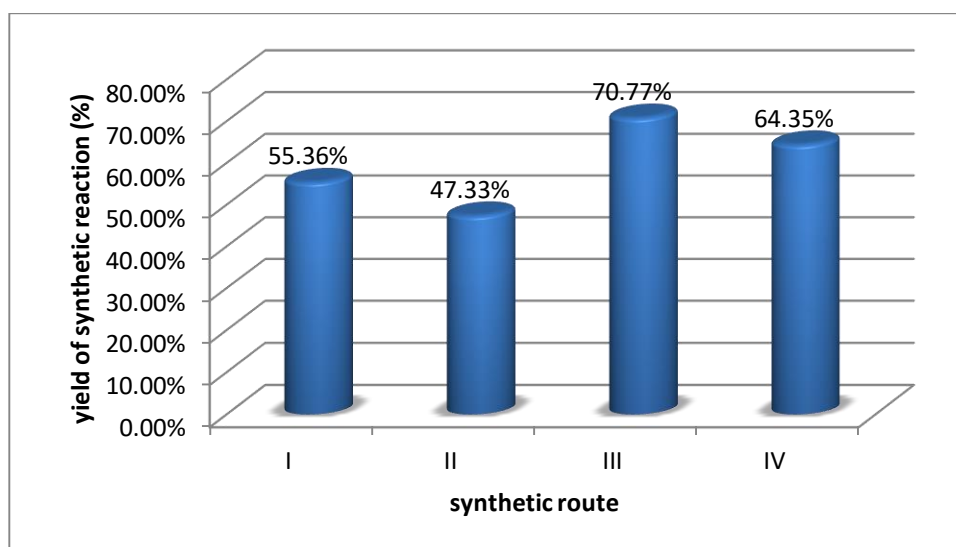


Figure 7: Graphic values of yield reactions for four different acetaminophen synthesis

The results of FTIR analysis (**Fig. 8.**) of crude acetaminophen for four different methods of laboratory synthesis indicate that in addition to the largest yield percentage of synthetic reactions, by using the third (III) method of synthesis of molecules we get acetaminophen with a very high degree of purity, which 99.20% cent overlaps with the pure paracetamol (European Pharmacopoeia). We have identical case with the fourth method of synthesis, while FTIR characterization of samples of the first and second method of synthesis shows lower values and requires purification.

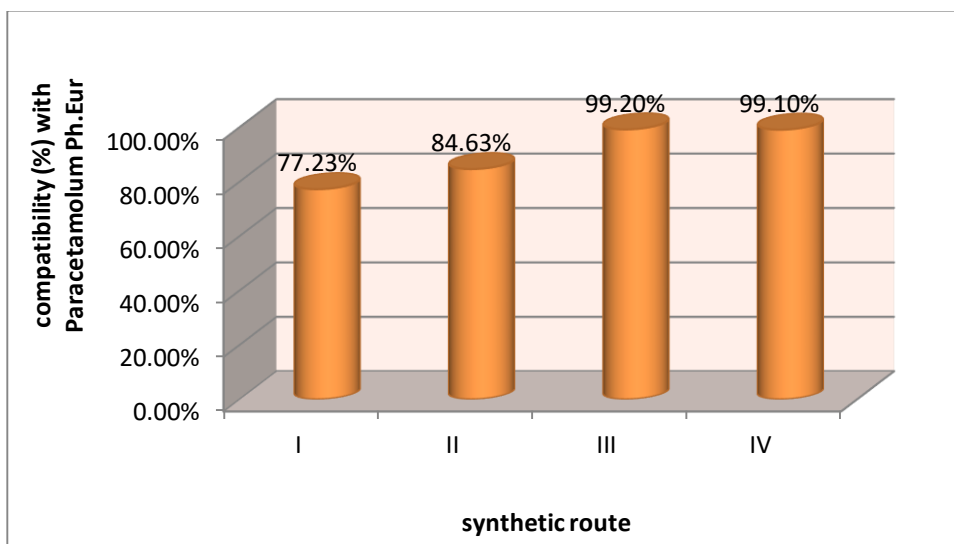


Figure 8: FTIR characterization synthesized samples of crude acetaminophen

The melting point of acetaminophen ranges from 168 to 172°C. **Figure 9** graphically shows the mean value of the melting point (°C) of crude acetaminophen for four different laboratory synthesis methods which are in accordance with data for Paracetamol Ph.Eur ¹⁶.

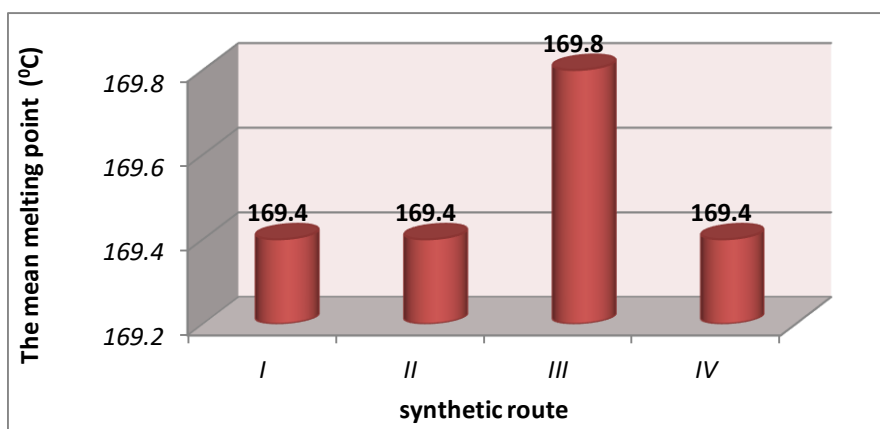


Figure 9: The Mean melting point (°C) of crude acetaminophen for 4 different synthesis methods

Monitoring of the flow of acetaminophen synthesis has been made by thin layer chromatography (**Fig. 10.**). We analyzed samples of p-aminophenol ($R_f = 0.12$) which was a starting component for the preparation of acetaminophen, then Paracetamol Ph.Eur and synthesized crude acetaminophen. Based on the obtained values of R_f and visual determination under ultraviolet light at 254 nm, we can see that the difference between R_f values of standard sample -Paracetamol Eur. Pharmacopoeia ($R_f=0.65$) and synthesized raw sample of acetaminophen ($R_f = 0.67$). By using the first method of synthesis, it pointed to the fact that the same should be additionally purified.

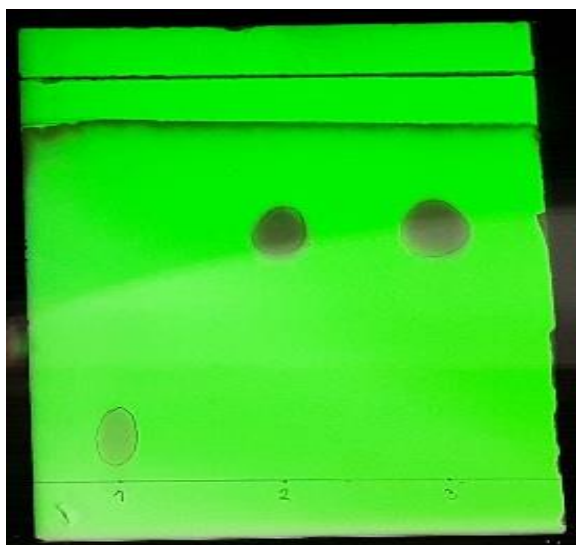


Figure 10: TLC chromatographic plate with samples 1) p-aminophenol 2) Paracetamol Ph.Eur. 3) the crude synthesized (synthesis method I) acetaminophen

As the solvent in the recrystallization of the crude acetaminophen phase in the first and second method of the laboratory synthesis aside from water we used C₁-C₃ alcohols. From the point of yield of recrystallization phase, as the preferred solvent we can use concentrated methanol (**Fig. 11.**). With the increasing number of carbon atoms in the structure of alcohol it reduces the solubility of paracetamol and increases the time required for crystallization.

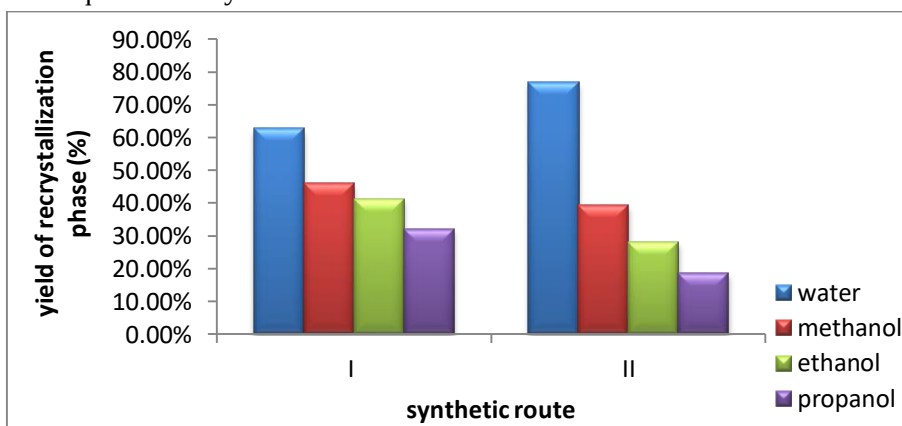


Figure 11: Yield of the crude acetaminophen recrystallization phase for first and second laboratory synthesis method.

As to the first and second laboratory method of synthesis of acetaminophen the effect of purification process with concentrated solvents is evident, because the FTIR analysis (**Fig. 12.**) demonstrated an increase in the degree of stringency as 99.4% and 99.5% compared to 77.23 % and 84.63% for the crude acetaminophen obtained by the first and second method of synthesis.

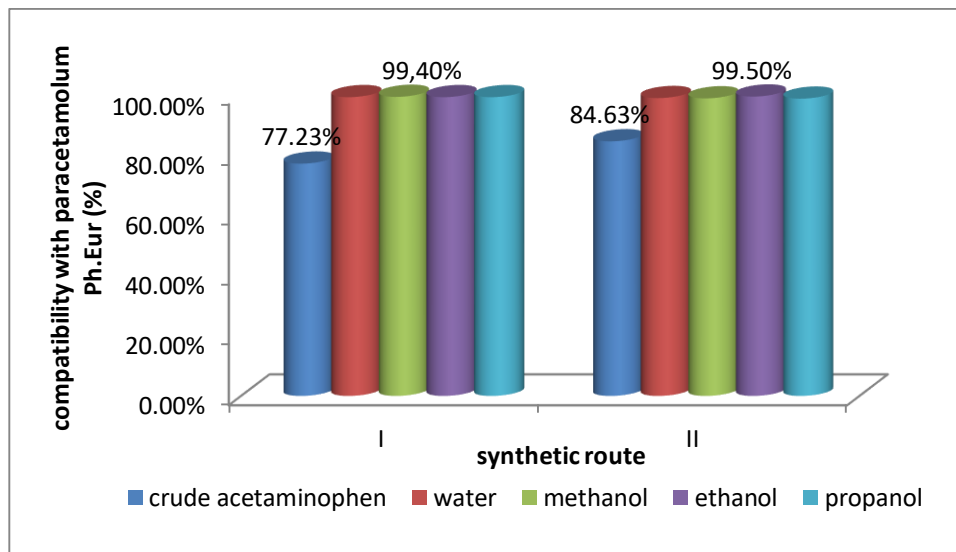


Figure 12: The influence of recrystallization on the purity of acetaminophen from the first two laboratory synthesis methods

The values of the melting point of samples of acetaminophen after recrystallization with concentrated solvents (**Table 1**) have increased and got closer to the upper limit of 172°C as opposed to the value of the melting points of raw samples before treatment (169,4°C), which supports the results of FTIR characterization.

Table-1: Melting points of recrystallized acetaminophen sample I and II laboratory synthesis method

Solvents	I Laboratory Method Of Synthesis	Ii Laboratory Method Of Synthesis
	the mean melting point (°C)	the mean melting point (°C)
Water	169,7	169,6
Methanol	171,8	171,4
Ethanol	171,7	171,7
Propanol	171,6	171,7

We studied the effect of the aqueous binary mixtures of methanol concentration range of 20-80% on the yield of recrystallization phase of the crude acetaminophen along with optimization of crystallization conditions at room temperature and at a temperature of 6°C for 24 h (**Figure 13**). It is important to point out that lowering the temperature of crystallization goes in favor of increasing the yield of

recrystallization phase. If we use 60% methanol as a medium of crystallization by cooling the sample at 6°C for 24 h, we achieve 37.8% higher yield of the crystallization phase relative to room temperature conditions. Increase of yield of the crystallization when using 80% methanol as a medium, is 88.3% in stated conditions.

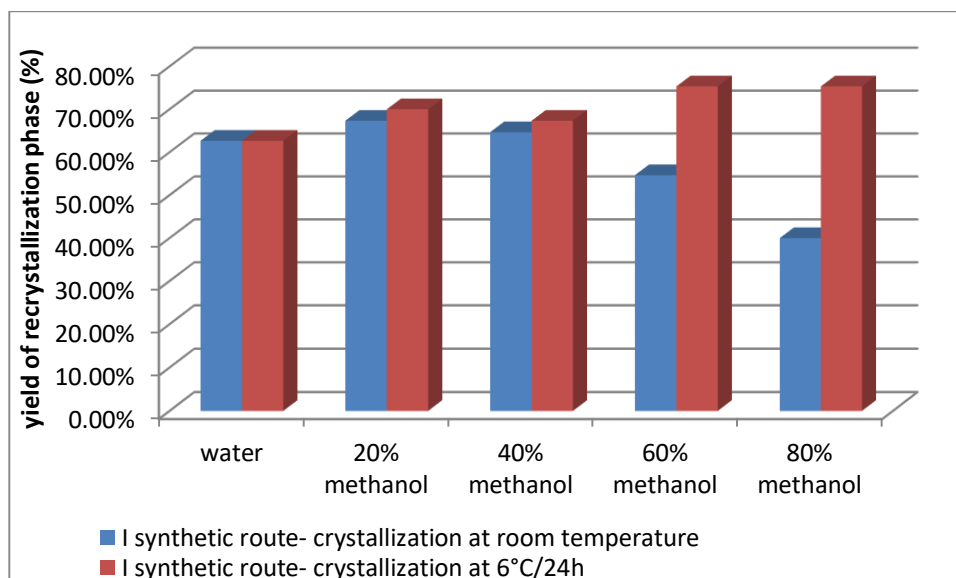


Figure 13: The effect of binary mixtures to yield of acetaminophen crystallization under different conditions

FTIR analysis of purified samples of acetaminophen under the conditions of crystallization at room temperature and at a temperature of 6°C for 24 h (**Figure 14**) shows that the binary mixture of methanol is suitable as crystallization medium to acetaminophen because they have reached an adequate percentage of correspondence (from 97.83 to 98, 83%) of given samples with Paracetamol Ph.Eur.

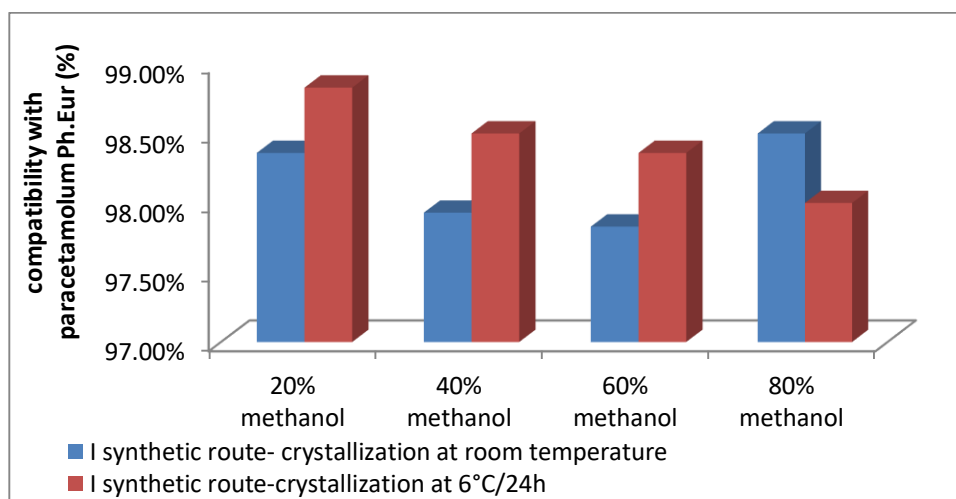


Figure 14: FTIR analysis of acetaminophen samples crystallized at room temperature and at 60°C for 24^h

The complexing of acetaminophen with solutions of beta cyclodextrin, and their impact on the solubility of a given compound, has been used in the study of phase solubility. The complexing has been also used to determine the type of chart of phase solubility. **Figure 15** presents the diagram of phase solubility of synthesized acetaminophen in which we can observe that the solubility of acetaminophen increases linearly with increasing concentration of beta-cyclodextrin (0,002-0,01M) which shows the characteristic phase of A_L solubility curve. According to Higuchi and Connors¹⁴, provided A_L solubility curve indicates the formation of complexes between the substrate (acetaminophen) and ligand (beta-cyclodextrin), depending on the relative concentration of cyclodextrin. All this indicates that the use of a 0.01 M aqueous solution of beta-cyclodextrin increases the solubility of acetaminophen to 13.22% compared to its solubility in water. The results of ANOVA analysis with p-value of 3,995-08 show that there is obvious statistical difference at the level of significance of 0.05 between increasing concentrations of aqueous solutions of cyclodextrin, and the concentration of synthesized acetaminophen.

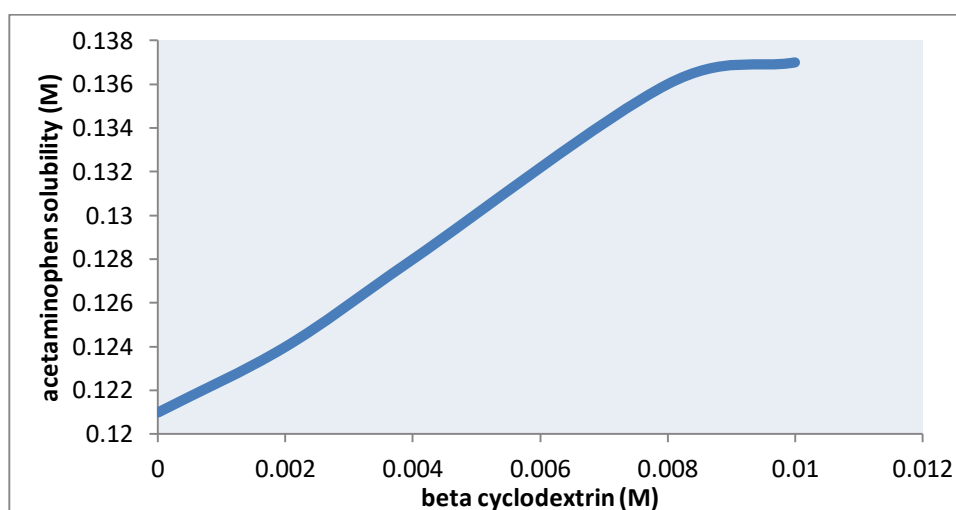


Figure 15: Diagram of phase solubility synthesized acetaminophen

CONCLUSION

On the basis of these results, we may conclude that the synthesis of acetaminophen is effective and acceptable; thanks to the wide range of use in different inflammatory conditions it is understandable why it is the main active ingredient in preparations against fever. By applying four different methods of synthesis of acetaminophen, the significant progress has been made in terms of percentage yield of synthetic reactions after the third (III) method of synthesis, without the presence of a catalyst (H_3PO_4) while warming under reflux of 70.77%. The results of FTIR characterization of samples obtained by the third method of synthesis suggest that produced molecules of acetaminophen have considerable degree of purity, which 99.20% cent overlaps with the pure acetaminophen (European Pharmacopoeia). Physicochemical analysis of the melting point of acetaminophen samples synthesized in four different methods are in accordance with literature values in the range from 168 to 172°C. During the process of recrystallization of acetaminophen synthesized samples it has been found that with increasing number of carbon atoms in the structure of the alcohol decreases the solubility of acetaminophen and increases the time required for crystallization. Analysis of the results of UV spectroscopy points out the possibility of the complexing of acetaminophen with cyclodextrin, because obtained linear dependence of increased

acetaminophen solubility with the increase of beta-cyclodextrin concentrations indicates a potential interaction.

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REFERENCES

1. United States Pharmacopeia and National Formulary (USP 29 NF 24). Supplement No. 2. Rockville, 2006, 3711.
2. J.R. Bales, J.K. Nicholson, P.J. Sadler, Clin. Chem. 1985, 31, 757.
3. H. M. Zaki, Spectroscopy Surface Analysis of Paracetamol and Paracetamol and Excipient Systems, Doctoral Dissertation, University of Manchester, Faculty of Engineering and Physical Sciences, 2011.
4. L. Kalantzi, C. Reppas, J.B. Dressman, G.L. Amidon, H.E. Junginger, K.K. Midha, V.P. Shah, S.A. Stavchansky, D. M. Barends, J.Pharm. Sci., 2006, 95, 4.
5. Budavari S, The Merck Index: an encyclopedia of chemicals, drugs and biologicals, 12th ed. Rahway, New Jersey, Merck and Co., Inc. 1996.
6. L.R. Agnew, D. L. Cruickshank, T. McGlone, C. C. Wilson, Chem. Commun., 2016, **52**, 7368.
7. C. Valder, D. Merrifield, Pharmaceutical Technology. SmithKline Beecham R&D News 1996.
8. N. Blagden, M. De Matas, P.T. Gavan, P. York, Adv. Drug Deliv. Rev., 2007, 59, 617.
9. B. Y Shekunov, P. York, J. Cryst. Growth. 2000, 211, 122.
10. H. Tung, E.L. Paul, M. Midler, J.A. McCauly, Crystallization of organic compounds: an industrial perspective, New Jersey, John Wiley & Sons, 2009.
11. F. Požgan, B.Štefane, Uvod v laboratorijsko organsko kemijo, Ljubljana, Fakulteta za kemijo in kemijsko tehnologijo, Univerza v Ljubljani, 2011.
12. A.S. Balte, P.K.Goyal, S.P.Geji, J.Chem.Pharm. Res. 2012, 4, 2391.
13. M.E. Davis, M. E. Brewster, Nat. Rev. Drug Discov. 2004, 3, 1023.
14. T. Higuchi, K.A. Connors, Adv.Anal.Chem.Instrum. 1965, 4, 117.
15. M. F. Striegel, J. Hill, Thin-Layer Chromatography for Binding Media Analysis. The Getty Conservation Institute, Los Angeles 1996.
16. W.M. Haynes, CRC handbook of chemistry and physics. 94th edition. CRC Press LLC, Boca Raton, 2013-2014, 3-314.

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